

MTH310/MTH520: Graph Theory (GT)

Quiz 1

January 24, 2025

Solve either one of Q1 or Q2 in your answer booklet. In case both are attempted, the answer with the maximum marks will be taken as the final score.

1. You have already seen triangle-free graphs (graphs that do not contain a $\triangle$) in Tut 1. Now that you are further aware of graph terminology and concepts, let's give a better characterization of triangle-free graphs:

- a) Can you give a lower bound on $g(G)$ of a triangle-free graph G ?
- b) Can you give an upper bound on the clique number $\omega(G)$ (order of largest complete subgraph) of a triangle-free graph G ?
- c) Prove that G is triangle-free if and only if $N(v)$ is an independent set $\forall v \in V(G)$.
- d) Now let G be triangle free and $v \in V(G)$ be a vertex of maximum degree. Show that $\sum_{u \in N(v)} d(u) \leq \Delta(G)(n - \Delta(G))$.
- e) Prove that G of order n is triangle-free implies that $|E(G)| \leq \frac{n^2}{4}$.

(Hint: Use part d) and $2|E(G)| = \sum_{v \in V(G)} d(v)$.)

[1+1+2+3+3=10 points]

OR

2. Consider the family of hypercube graphs Q_n which is formed by the vertices and edges of a n -dimensional hypercube where $n \in \mathbb{N}$. More formally, $Q_n = (\{0, 1\}^n, \{\{u, v\} \mid u, v \in \{0, 1\}^n \text{ and } u \text{ and } v \text{ differ by exactly one coordinate}\})$. For some $n \in \mathbb{N}$,

- a) Is Q_n regular?
- b) Show that $\forall \{u, v\} \subseteq 2^{V(Q_n)}$, $d_{Q_n}(u, v)$ equals the exact number of coordinates that u and v differ in.
- c) Show that $\forall v \in V(Q_n), \exists! u \in V(Q_n), d_{Q_n}(v, u) = \max_{x \in V(Q_n)} d_{Q_n}(v, x) = \text{diam}(Q_n)$.
- d) For $n \geq 2$, prove that Q_n contains a subgraph $H \cong Q_n \setminus H$.

(Hint: Observe that the number of vertices with first coordinate zero, is exactly the same as the number of vertices with first coordinate one.)

- e) Prove using part d) that $\text{circ}(Q_n) = 2^n$.

[0.5+1.5+1.5+3+3.5=10 points]